Yu Chen

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Education

Tsinghua University Institute for Interdisciplinary Information Sciences 2023 – 2028 (expected)
Ph.D. Candidate in Computer Science. Advisor: Prof. Longbo Huang.

Tsinghua University Department of Mathematics 2019 – 2023
Bachelor of Science in Mathematics.

Research Interests

My research lies at the intersection of AI and decision-making: I develop the theory of sequential decision-making under uncertainty, from bandits to reinforcement learning, and study how AI systems such as LLM agents make decisions. My current work follows three directions: decision-making in LLMs and LLM agents, heavy-tailed online learning, and risk-sensitive reinforcement learning; in each, I look for formulations that lead to principled algorithms and, where possible, provable guarantees.

Selected Publications

(* indicates equal contribution; reverse chronological within each category.)

Preprints

[arXiv'26] Ruishuo Chen, Weijia Li, Xun Wang, **Yu Chen**, Leheng Cai, and Longbo Huang. "One Proposal for Every Margin: Zero-Shot Amortized Sequential Importance Sampling for Binary Matrices." arXiv preprint, 2026.

[arXiv'26] Ruishuo Chen, Xun Wang, **Yu Chen**, Zhuoran Li, and Longbo Huang. "The Router Within: Eliciting Native Skill Routing from a Frozen LLM." arXiv preprint, 2026.

[arXiv'26] **Yu Chen***, Ruishuo Chen*, Xun Wang, Zhuoran Li, and Longbo Huang. "Optimal Skill Selection for LLM Agents with Provable Bicriteria Guarantees." arXiv preprint, 2026.

[arXiv'26] Xun Wang, Ruishuo Chen, Zhuoran Li, **Yu Chen**, and Longbo Huang. "When Context Returns: Toward Robust Internalization in On-Policy Distillation." arXiv preprint, 2026.

Conference and Journal Publications

[UAI'26] **Yu Chen***, Siwei Wang*, Longbo Huang, and Wei Chen. "On the Sublinear Regret of Continuous K-Max Bandits." Forty-second Conference on Uncertainty in Artificial Intelligence, 2026.

[ICML'26] **Yu Chen**, Yuhao Liu, Jiatai Huang, Yihan Du, and Longbo Huang. "Best-of-Both-Worlds for Heavy-Tailed Markov Decision Processes." Forty-third International Conference on Machine Learning, 2026.

[ICML'26] Ruishuo Chen, **Yu Chen**, Zhuoran Li, and Longbo Huang. "PowerFlow: Unlocking the Dual Nature of LLMs via Principled Distribution Matching." Forty-third International Conference on Machine Learning, 2026. [Code]

[ICML'26] Rui Hu, **Yu Chen**, and Longbo Huang. "Finite-time Convergence Analysis of Actor-Critic with Evolving Reward." Forty-third International Conference on Machine Learning, 2026.

[ICLR'26] Yuhao Liu, **Yu Chen**, Rui Hu, and Longbo Huang. "Finite-Time Convergence Analysis of ODE-based Generative Models for Stochastic Interpolants." The Fourteenth International Conference on Learning Representations, 2026.

[ICML'25] Yuhao Liu, **Yu Chen**, Rui Hu, and Longbo Huang. "Finite-Time Analysis of Discrete-Time Stochastic Interpolants." Forty-second International Conference on Machine Learning, 2025.

[ICLR'25] **Yu Chen***, Jiatai Huang*, Yan Dai*, and Longbo Huang. "uniINF: Best-of-Both-Worlds Algorithm for Parameter-Free Heavy-Tailed MABs." The Thirteenth International Conference on Learning Representations, 2025. Spotlight (Top 5%).

[ICML'24] **Yu Chen***, Xiangcheng Zhang*, Siwei Wang, and Longbo Huang. "Provable Risk-Sensitive Distributional Reinforcement Learning with General Function Approximation." Forty-first International Conference on Machine Learning, 2024.

[ICML'24] Tonghe Zhang*, **Yu Chen***, and Longbo Huang. "Provably Efficient Partially Observable Risk-Sensitive Reinforcement Learning with Hindsight Observation." Forty-first International Conference on Machine Learning, 2024.

[TON'24] Pihe Hu, **Yu Chen**, Ling Pan, Zhixuan Fang, Fu Xiao, and Longbo Huang. "Multi-User Delay-Constrained Scheduling With Deep Recurrent Reinforcement Learning." IEEE/ACM Transactions on Networking, 2024.

[ICLR'24] **Yu Chen**, Yihan Du, Pihe Hu, Siwei Wang, Desheng Wu, and Longbo Huang. "Provably Efficient Iterated CVaR Reinforcement Learning with Function Approximation and Human Feedback." The Twelfth International Conference on Learning Representations, 2024.

[ICLR'23] Pihe Hu*, **Yu Chen***, and Longbo Huang. "Towards Minimax Optimal Reward-free Reinforcement Learning in Linear MDPs." The Eleventh International Conference on Learning Representations, 2023.

[MobiHoc'22] Pihe Hu, Ling Pan, **Yu Chen**, Zhixuan Fang, and Longbo Huang. "Effective multi-user delay-constrained scheduling with deep recurrent reinforcement learning." Twenty-third International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing, 2022.

[ICML'22] Pihe Hu, **Yu Chen**, and Longbo Huang. "Nearly Minimax Optimal Reinforcement Learning with Linear Function Approximation." Thirty-ninth International Conference on Machine Learning, 2022.

Selected Honors and Awards

Research Awards

- Spotlight (Top 5%), 13th International Conference on Learning Representations (ICLR), for "uniINF: Best-of-Both-Worlds Algorithm for Parameter-Free Heavy-Tailed MABs." 2025

Fellowship and Scholarship

- China National Scholarship, top scholarship in China for graduate students (2% domestically). 2024
- China National Scholarship, top scholarship in China for undergraduate students (2% domestically). 2022
- Xiao Qin Scholarship, recognition for the best student of each class in Department of Mathematics (3 / 111), Tsinghua. 2022
- Tsinghua University Scholarship, recognition for academic performance. 2020, 2025
- Tsinghua University Scholarship, recognition for social work performance. 2020, 2021

Competitions

- First Prize (Beijing Area), Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM). 2020
- First Prize, Chinese Mathematical Olympiad in Provinces. 2018
- First Prize, Chinese Mathematical Olympiad in Provinces. 2017
- First Prize, National Olympiad in Informatics in Provinces. 2017

Academic Services

Conference Reviewing. NeurIPS, ICLR, ICML, AISTATS, AAAI.

Journal Reviewing. IEEE Transactions on Mobile Computing, Machine Learning, Expert Systems with Applications.